

## Project Title: University Management System

### I. Project Description:

The aim of this project is to design and implement a comprehensive University Management System that facilitates the management of various academic and administrative tasks within a university environment. The system will provide functionalities for managing students, courses, instructors, departments, enrollment, grading, scheduling, and other essential aspects of university operations.

### II. Project Requirements:

#### 1. User Management:

- ◆ Implement user authentication and authorization mechanisms to distinguish between different user roles (e.g., administrators, faculty, students).
- ◆ Allow users to register, login, and manage their profiles.

#### 2. Student Management:

- ◆ Enable registration of new students with relevant personal information.
- ◆ Allow students to enroll in courses, view their academic records, grades, and course schedules.
- ◆ Implement functionalities for academic advising and course recommendations.

#### 3. Course Management:

- ◆ Provide functionalities for creating, updating, and deleting courses.
- ◆ Associate courses with departments and instructors.
- ◆ Define prerequisites for courses and enforce them during enrollment.

#### 4. Instructor Management:

- ◆ Enable registration of new instructors with relevant personal and departmental information.
- ◆ Allow instructors to manage their teaching assignments, view student

rosters, and submit grades.

#### 5. Department Management:

- ◆ Implement functionalities for creating and managing departments, including budgeting and staffing.
- ◆ Associate departments with courses and instructors.

#### 6. Enrollment and Grading:

- ◆ Implement course enrollment functionalities for students, including adding and dropping courses.
- ◆ Allow instructors to submit grades for enrolled students.

#### 7. Scheduling and Time Slots:

- ◆ Provide functionalities for managing course schedules, including defining time slots and room assignments.
- ◆ Ensure that scheduling constraints are enforced to avoid conflicts.

#### 8. Reporting and Analytics:

- ◆ Generate reports on enrollment statistics, course performance, student progress, etc.
- ◆ Implement data visualization techniques to present insights from the collected data.

### III. Project Deliverables:

1. Database Design: Design a relational database schema based on the provided project requirements. Include entity-relationship diagrams (ERD), table definitions, primary and foreign key constraints, and any additional database features required.
2. Application Development: Implement the university management system using appropriate technologies and frameworks such as .NET Framework, C#, and MS-SQL. Develop user interfaces for interacting with the system, including forms for data entry, views for displaying information, and interactive dashboards for analytics.
3. Documentation: Provide comprehensive documentation covering system architecture, database schema, user guides, and development guidelines.

Include explanations of design decisions, implementation details, and instructions for setting up and running the application.

4. Presentation: Prepare a presentation to demonstrate the functionalities of the university management system. Highlight key features, technical aspects, and challenges faced during the development process.

IV. Project Timeline:

- Week 1-2: Requirement analysis and database design
- Week 3-5: Application development and testing
- Week 6: Documentation and finalizing project deliverables
- Week 7: Presentation and project submission

V. Project Evaluation:

The project will be evaluated based on the following criteria:

- Fulfillment of project requirements
- Database design and normalization
- Implementation of functionalities and user interfaces
- Documentation quality and completeness
- Presentation quality and clarity

**\*\*Note:\*\*** Students are encouraged to seek guidance from the instructor and utilize available resources such as textbooks, online tutorials, and development tools to successfully complete the project.